

$$A+B=C$$

Reinforcement Learning (RL)

$$\sqrt{a}$$

$$2^{-}$$





Reinforcement Learning (RL)

- An agent interacts with an environment.
- At each step: state (s) $\rightarrow$ action (a) $\rightarrow$ new state + reward.
- Goal: Learn how to act to maximize long-term rewards.



A $P = C$

Program	Agent	Environment	Action	State
A	Control chip (processor)	Motors & sensors	Electrical signals	Motor positions, sensor readings (dust detected, battery level)
B	Robot vacuum	Room layout (walls, furniture)	Wheel and brush movements	Robot's internal map of obstacles and cleaned/uncleaned areas
C	Smart home system	Entire house	Robot's navigation commands	House state (which rooms are clean/dirty, charging dock status)





Model-Based RL

- Agent tries to understand how the environment works.
- Builds/uses a transition model: $P(s' | s, a)$, $R(s, a)$.
- Can be given (rules of chess) or learned (robot physics).
- Advantages: Can plan, efficient learning, handles partial observability.
- Disadvantages: Hard to build accurate model in complex environments.





Model-Free RL

- Agent does not build a model of the environment.
- Directly learns what actions work best.
- Types:
 1. Action-Utility Learning (Q-learning): learns $Q(s,a)$.
 2. Policy Search: learns $\pi(s)$ directly.
- Advantages: Simpler, no model needed.
- Disadvantages: Can be sample-inefficient (needs more trial and error).





Passive vs Active RL

- Passive RL:
 - Fixed policy (agent already knows actions).
 - Goal = learn utilities of states under that policy.
- Active RL:
 - Policy is not fixed.
 - Learns both which actions lead to rewards and which states are valuable.
 - Challenge: Exploration vs exploitation.





Markov Decision Process (MDP)

- $S_0, A_0, R_0, S_1, A_1, R_1, S_2, A_2, R_2, S_3, A_3, R_3, \dots$
- Markov Property: State is depend on immediate previous state.
- Policy

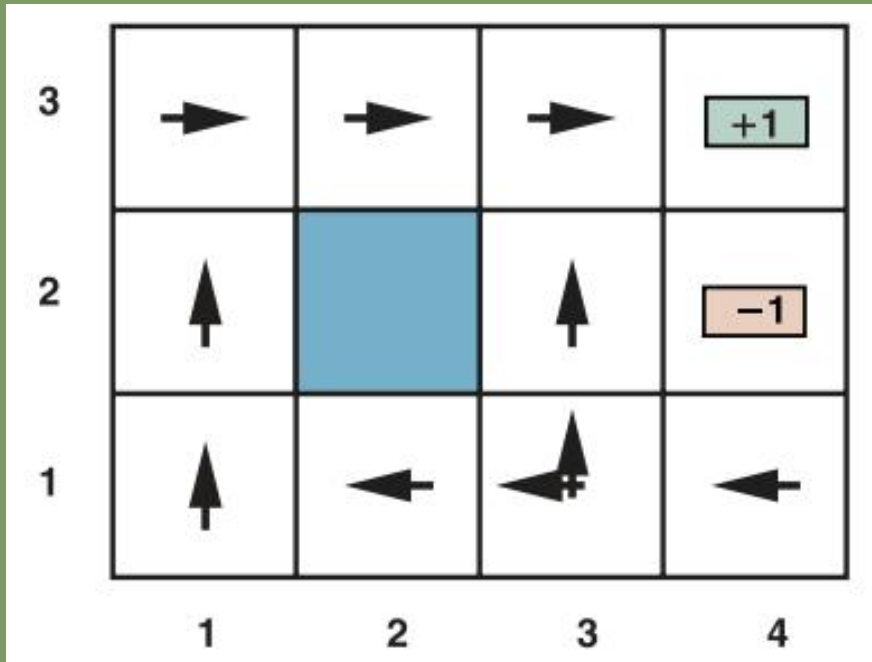
$\pi(a|s)$ =Probability

- Return $G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \gamma^3 R_{T+3}$
where, γ is discount factor $0 \leq \gamma \leq 1$
- Goal: Find the policy which maximizes return





Grid Example

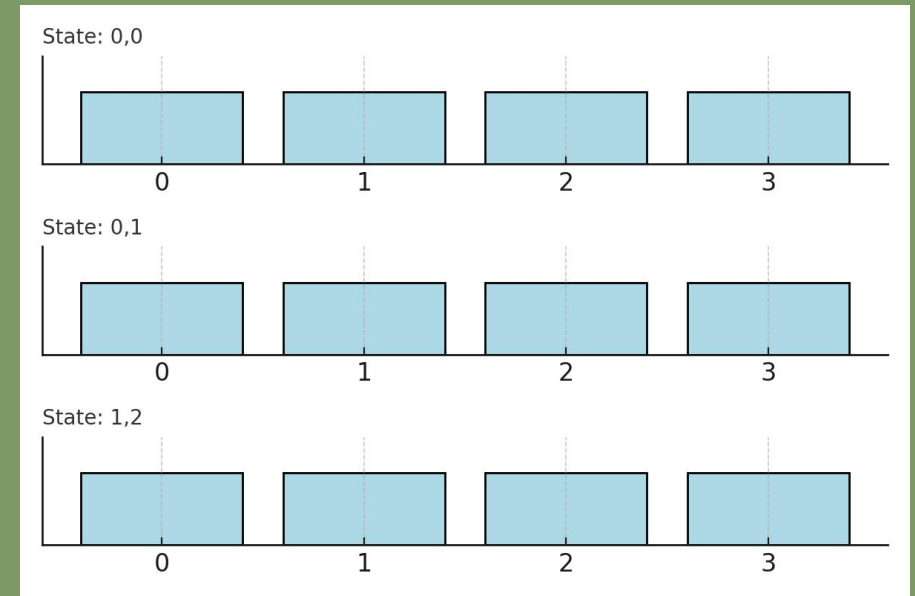


- Episodes
- World Model
- $p(S', r | s, a)$



Example of Trajectory

State	Action	Next State	Reward	Return
0,1	2	1,1	-1	
1,1	3	1,2	-1	
1,2	1	2,2	-1	
2,2	0	2,3	-1	
2,3	0	3,3	-1	
3,3	1	3,2	-1	
3,2	2	2,2	-1	
2,2	2	2,1	1	-1.8
2,1	3	1,1	1	-1
1,1	1	0,1	0	0

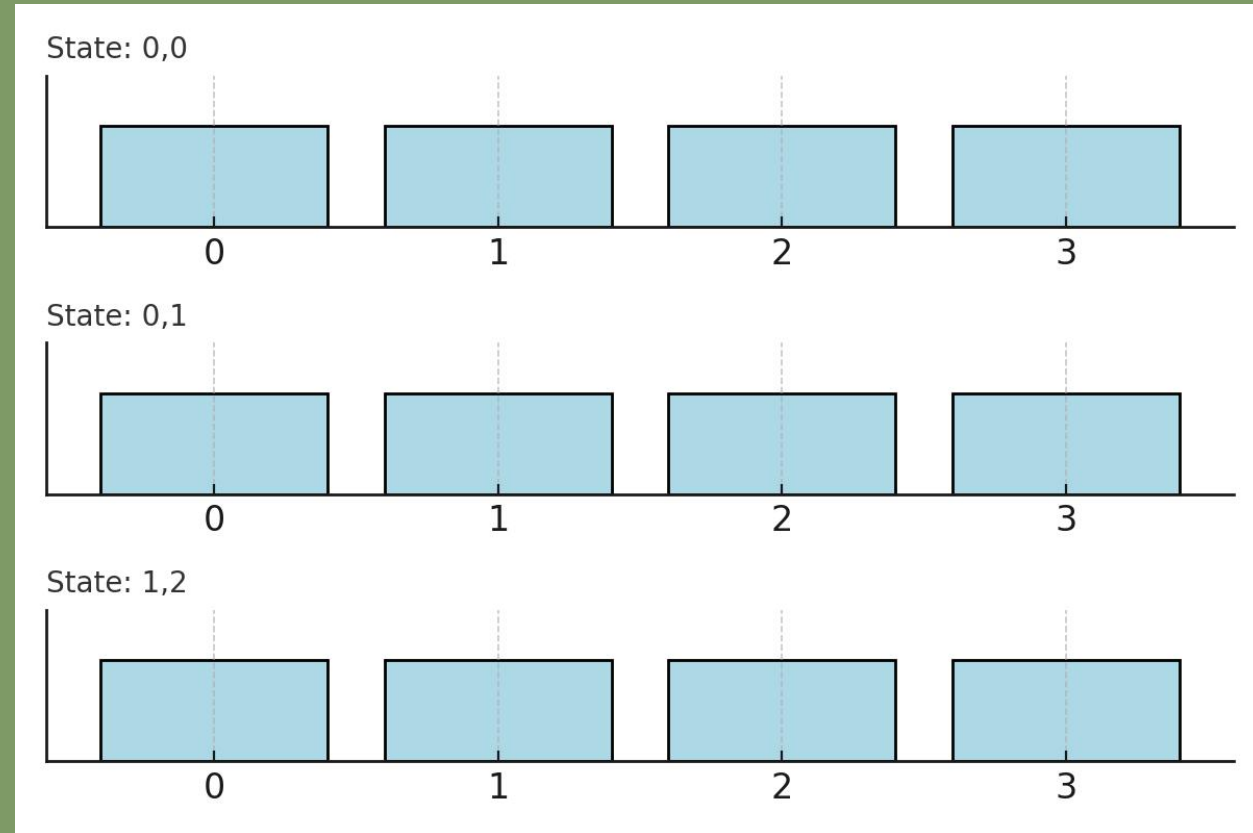


Policy

$\pi(a|s)$ = Probability

Return $G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \gamma^3 R_{T+3}$

Policy Gradient-on good and bad action





Value Function

- Keep track of Average Return (G_t) Expected when following a certain policy π in a certain state (s) or state+action (s,a)
- State value function $V_\pi(S)$
- Action value function $Q_\pi(S,A)$





Generalized Policy Iteration

1) Policy Gradient

- dependent on policy Function

2) Policy Gradient method, with value function

- Balance of policy and value function

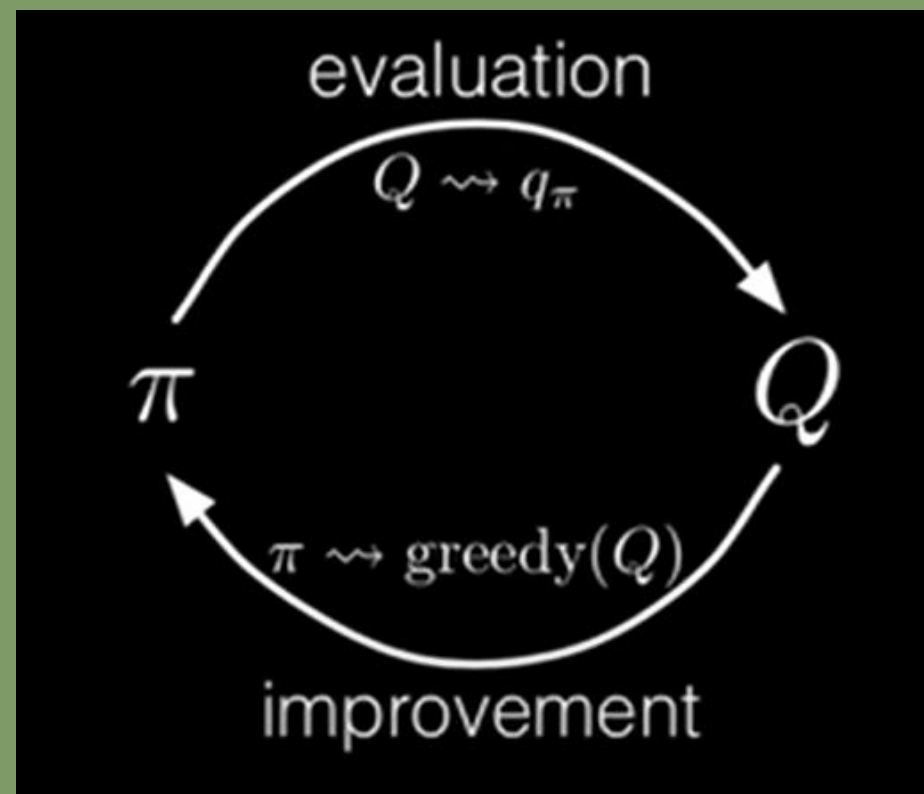
3) Just use action -value (Q) function

- dependent on value function





Generalized Policy Iteration





Evaluation

	0	1	2	3	4
0	→	→	→	→	↓
1	↑	↑	↑	↓	↓
2	↑	↑	↑	→	↑

	0	1	2	3	4
0	0: -3.65 1: -3.86 2: -3.63 3: -3.36	0: -3.36 1: -3.68 2: -3.67 3: -2.95	0: -2.95 1: -2.94 2: -3.36 3: -2.44	0: -2.39 1: -1.80 2: -2.94 3: -1.80	0: -1.74 1: -1.00 2: -2.28 3: -1.59
1	0: -3.69 1: -4.12 2: -3.92 3: -3.69	0: -3.36 1: -3.92 2: -3.91 3: -3.68	0: 0.00 1: 0.00 2: 0.00 3: 0.00	0: -2.38 1: -1.00 2: -1.77 3: -1.00	0: -1.62 1: 0.00 2: -1.73 3: -0.96
2	0: -3.95 1: -4.14 2: -4.14 3: -3.95	0: -3.69 1: -3.87 2: -4.05 3: -3.88	0: 0.00 1: 0.00 2: 0.00 3: 0.00	0: -1.40 1: -0.88 2: -0.88 3: 0.00	0: 0.00 1: 0.00 2: 0.00 3: 0.00





Evaluation

	0	1	2	3	4
0	0: -3.65 1: -3.86 2: -3.63 3: -3.36	0: -3.36 1: -3.68 2: -3.67 3: -2.95	0: -2.95 1: -2.94 2: -3.36 3: -2.44	0: -2.39 1: -1.80 2: -2.94 3: -1.80	0: -1.74 1: -1.00 2: -2.28 3: -1.59
1	0: -3.69 1: -4.12 2: -3.92 3: -3.69	0: -3.36 1: -3.92 2: -3.91 3: -3.68	0: 0.00 1: 0.00 2: 0.00 3: 0.00	0: -2.38 1: -1.00 2: -1.77 3: -1.00	0: -1.62 1: 0.00 2: -1.73 3: -0.96
2	0: -3.95 1: -4.14 2: -4.14 3: -3.95	0: -3.69 1: -3.87 2: -4.05 3: -3.88	0: 0.00 1: 0.00 2: 0.00 3: 0.00	0: -1.40 1: -0.88 2: -0.88 3: 0.00	0: 0.00 1: 0.00 2: 0.00 3: 0.00

Q function estimate eventually goes from...

approximating $Q_\pi \rightarrow$ approximating Q_*

policy π , based on Q , approaches π_* (optimal policy)





Exploration and Exploitation

Exploration:

Means trying new actions to discover their effects.

Goal: Gather information about the environment.

Exploitation:

Means choosing the best-known action based on past experience.

Goal: Maximize immediate reward using what is already known.





Exploration and Exploitation Trade-off

How RL Handles It

ϵ -greedy method:

With probability $\epsilon \rightarrow$ explore (choose random action).

With probability $1-\epsilon \rightarrow$ exploit (choose best-known action).

Example: $\epsilon = 0.1 \rightarrow$ 10% of the time random, 90% best action.

